Multi-Client PMO Portfolio Tracker

Methodology & Documentation Report

A PostgreSQL + Power BI Portfolio Project Simulating Multi-Client IT Services Delivery

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3. Executive Summary

This project simulates a Project Management Office (PMO) supporting a multi-client IT services delivery organization. It was built to demonstrate an end-to-end data pipeline — from relational schema design through SQL analysis to an interactive executive dashboard — using realistic, portfolio-scale synthetic data rather than a toy or single-table exercise.

At a glance:

- 15 concurrent projects across 10 clients and 7 delivery teams (ERP, Cloud Migration, Cybersecurity, Data & Analytics, Digital Transformation, CRM Rollout, Infrastructure Modernization)
- A 5-table relational schema (Projects, Milestones, Resources, RiskLog, WeeklyStatus) built in PostgreSQL
- 111 milestones, 68 resource allocations, 37 logged risks, and 150 weekly status snapshots
- A 10-query SQL showcase spanning beginner filtering through CTEs, window functions, and correlated subqueries
- A 4-page interactive Power BI dashboard with a full DAX measure library and drillthrough detail page

4. Business Problem & Context

An IT services PMO managing 15 concurrent client engagements across 7 delivery teams needs consolidated, real-time visibility into budget health, schedule adherence, risk exposure, and resource allocation. Without a centralized system, this visibility is typically assembled manually — a PMO Analyst compiling individual project statuses into a single view ahead of each Steering Committee meeting, a process that is slow, error-prone, and not repeatable on demand.

This project addresses that gap by building a governed, queryable data layer and a self-service reporting layer on top of it, so portfolio-wide questions can be answered in seconds rather than through manual compilation.

5. Objectives

- Identify which projects are at risk of budget overrun or schedule slippage before a client or executive raises it.
- Surface the portfolio's highest-severity open risks and track how the risk backlog is trending over time.
- Detect resource over-allocation across simultaneous project assignments.
- Provide an automated, rule-based escalation signal instead of relying on manual judgment calls.
- Make all of the above self-service through an interactive dashboard, not a static report.

6. Data Model & Schema Design

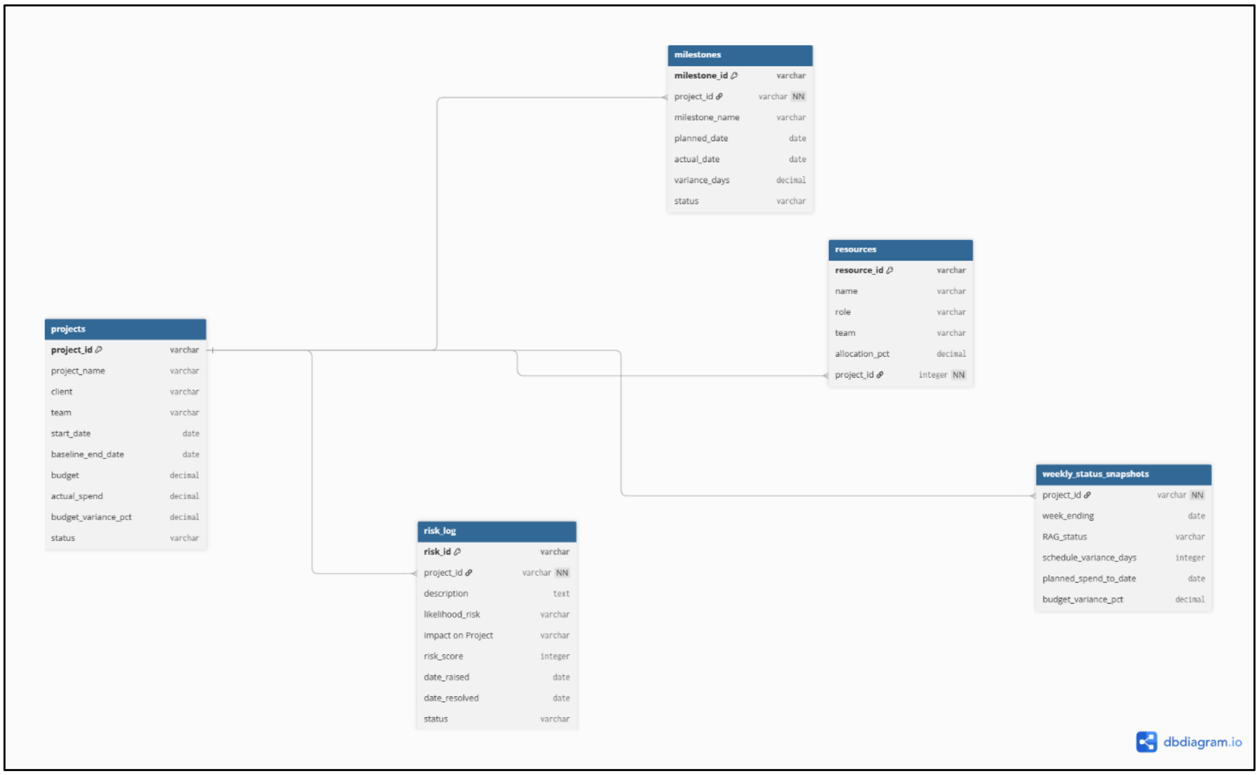


Table Overview

Table	Rows	Key Columns	Purpose
Projects	15	project_id (PK), client, team, status	Central project record — one row per engagement
Milestones	111	milestone_id (PK), project_id (FK)	Deliverable checkpoints per project
Resources	68	resource_id (PK), project_id (FK)	People staffed to projects and their allocation %
RiskLog	37	risk_id (PK), project_id (FK)	Identified risks, likelihood/impact/score
WeeklyStatus	150	project_id (FK), week_ending	Weekly time-series of RAG status and spend

Key Design Decisions

- Client and team fields were denormalized directly onto Projects and Resources rather than split into separate lookup tables, a deliberate simplicity trade-off appropriate for a portfolio-scale dataset, at the cost of some redundancy that a production system at larger scale would likely normalize.
- All four child tables relate to Projects with single-direction cross-filtering, keeping filter propagation predictable and avoiding ambiguous filter paths in the BI layer.
- risk_score is pre-computed from a standard likelihood × impact matrix (1–9 scale) rather than derived at query time, keeping risk prioritization consistent across both the SQL layer and the dashboard.

7. Tools & Technology Stack

Tool	Role in This Project
PostgreSQL	Primary relational database
DBeaver	Database client for import, querying, and schema management
Power BI Desktop	Data model, DAX measures, and the interactive dashboard
Excel	Used to build and validate the source dataset prior to loading into PostgreSQL.

8. Methodology / Build Process

The project followed a full data pipeline, in order:

1. Synthetic data generation — realistic multi-client, multi-team project data modeled in Excel
2. Schema design — 5-table relational structure, documented in DBML (dbdiagram.io) separate from the executable DDL
3. Database import — CSV data loaded into PostgreSQL via DBeaver
4. SQL validation — every query tested against real data (via SQLite simulation) before being finalized, catching data-type and formatting bugs early
5. BI layer, data model — relationships, and DAX measures built in Power BI, followed by page-by-page dashboard construction

9. SQL Analysis — Key Highlights

The full 10-query set spans four difficulty tiers, from basic filtering to correlated subqueries and forecasting logic. Highlights below; the complete set is included in the project repository.

Open Risk Count per Project (JOIN + GROUP BY)

Question: How many open risks is each project currently carrying? Uses a LEFT JOIN so projects with zero risks still appear with a count of 0, rather than being silently dropped.

```sql
SELECT
    p.project_id,
    p.project_name,
    COUNT(r.risk_id) AS open_risk_count
FROM projects p
LEFT JOIN risklog r
    ON p.project_id = r.project_id
    AND r.risk_state = 'Open'
GROUP BY p.project_id, p.project_name
ORDER BY open_risk_count DESC;
```

	project_id	project_name	open_risk_count
1	P008	Azure Cloud Migration - Phase 2	4
2	P011	Hybrid Cloud Backup Modernization	4
3	P012	Network & Data Center Refresh	4
4	P002	Customer Data Platform Rollout	3
5	P005	Zero Trust Security Rollout	3
6	P004	Guest Experience Mobile App	2
7	P010	Financial Close Automation	2
8	P001	Identity & Access Management Overhaul	2
9	P014	Contact Center CRM Migration	2
10	P003	SAP S/4HANA Migration	2
11	P007	BI & Reporting Modernization	1
12	P013	Claims Processing Automation	1
13	P006	Salesforce CRM Modernization	0
14	P015	Warehouse Management System Upgrade	0
15	P009	E-Commerce Platform Replatform	0

Top 3 Riskiest Projects per Team (CTE + RANK)

Question: Within each delivery team, which projects carry the highest total open-risk score? Uses a CTE feeding a RANK () OVER (PARTITION BY team ...) window function.

```
WITH project_risk AS (
    SELECT
        p.project_id,
        p.project_name,
        p.team,
        SUM(r.risk_score) AS total_risk_score
    FROM projects p
    JOIN risklog r ON p.project_id = r.project_id
    WHERE r.risk_state = 'Open'
    GROUP BY p.project_id, p.project_name, p.team
),
ranked AS (
    SELECT
        *,
        RANK() OVER (PARTITION BY team ORDER BY total_risk_score DESC) AS
risk_rank
    FROM project_risk
)
SELECT project_id, project_name, team, total_risk_score, risk_rank
FROM ranked
WHERE risk_rank <= 3
ORDER BY team, risk_rank;
```

	project_id	project_name	team	total_risk_score	risk_rank
1	P008	Azure Cloud Migration - Phase 2	Cloud Migration	15	1
2	P011	Hybrid Cloud Backup Modernization	Cloud Migration	15	1
3	P014	Contact Center CRM Migration	CRM Rollout	12	1
4	P005	Zero Trust Security Rollout	Cybersecurity	19	1
5	P001	Identity & Access Management Overhaul	Cybersecurity	10	2
6	P002	Customer Data Platform Rollout	Data & Analytics	18	1
7	P007	BI & Reporting Modernization	Data & Analytics	9	2
8	P004	Guest Experience Mobile App	Digital Transformation	12	1
9	P013	Claims Processing Automation	Digital Transformation	6	2
10	P010	Financial Close Automation	ERP Implementation	8	1
11	P003	SAP S/4HANA Migration	ERP Implementation	7	2
12	P012	Network & Data Center Refresh	Infrastructure Modernization	10	1

Projects Spending Above Their Team Average (Correlated Subquery)

Question: Which projects are spending more than the average actual spend of their own team? The subquery re-evaluates per outer row, correlated on team, a common interview probe for genuine understanding versus memorized syntax.

```
SELECT
    p1.project_id,
    p1.project_name,
    p1.team,
    p1.actual_spend
FROM projects p1
WHERE p1.actual_spend > (
    SELECT AVG(p2.actual_spend)
    FROM projects p2
    WHERE p2.team = p1.team
)
ORDER BY p1.team, p1.actual_spend DESC;
```

	project_id	project_name	team	actual_spend
1	P008	Azure Cloud Migration - Phase 2	Cloud Migration	527,900
2	P014	Contact Center CRM Migration	CRM Rollout	651,700
3	P005	Zero Trust Security Rollout	Cybersecurity	229,000
4	P002	Customer Data Platform Rollout	Data & Analytics	760,600
5	P013	Claims Processing Automation	Digital Transformation	746,300
6	P010	Financial Close Automation	ERP Implementation	536,500

Automated Escalation Rule Engine (Multi-table CASE Logic)

Question: Given status, budget variance, and open high-impact risks, what escalation action should each project trigger automatically? Replaces manual judgment calls with a deterministic, auditable rule.

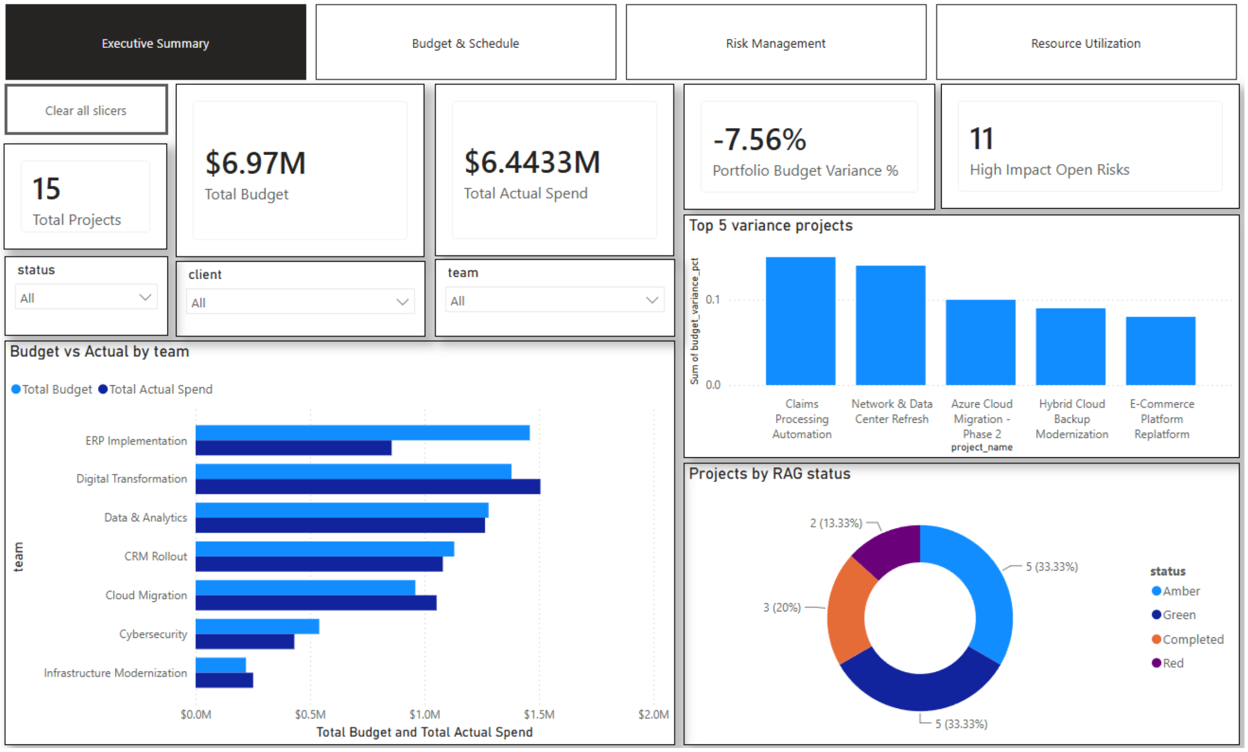
```
SELECT
    p.project_id,
    p.project_name,
    p.status,
    p.budget_variance_pct,
    COALESCE(risk.high_open_risks, 0) AS high_open_risks,
    CASE
        WHEN p.status = 'Red' AND COALESCE(risk.high_open_risks, 0) > 0
```

```
            THEN 'Escalate to Executive Sponsor'
        WHEN p.status = 'Red'
            THEN 'Escalate to PMO Director'
        WHEN p.status = 'Amber' AND p.budget_variance_pct > 5
            THEN 'Flag for Steering Committee'
        WHEN p.status = 'Amber'
            THEN 'Monitor'
        ELSE 'No Action'
    END AS escalation_action
FROM projects p
LEFT JOIN (
    SELECT project_id, COUNT(*) AS high_open_risks
    FROM risklog
    WHERE risk_state = 'Open' AND "impact on Project" = 'High'
    GROUP BY project_id
) risk ON p.project_id = risk.project_id
ORDER BY
    CASE p.status WHEN 'Red' THEN 1 WHEN 'Amber' THEN 2 ELSE 3 END;
```

	project_id	project_name	status	budget_variance_pct	high_open_risks	escalation_action
1	P012	Network & Data Center Refresh	Red	14	0	Escalate to PMO Director
2	P003	SAP S/4HANA Migration	Red	-22	1	Escalate to Executive Sponsor
3	P011	Hybrid Cloud Backup Moderniza	Amber	9	1	Flag for Steering Committee
4	P005	Zero Trust Security Rollout	Amber	4	1	Monitor
5	P013	Claims Processing Automation	Amber	15	0	Flag for Steering Committee
6	P002	Customer Data Platform Rollout	Amber	6	2	Flag for Steering Committee
7	P008	Azure Cloud Migration - Phase 2	Amber	10	1	Flag for Steering Committee
8	P001	Identity & Access Management (	Green	-37	0	No Action
9	P015	Warehouse Management Systen	Completed	0	0	No Action
10	P004	Guest Experience Mobile App	Green	1	2	No Action
11	P006	Salesforce CRM Modernization	Completed	4	0	No Action
12	P007	BI & Reporting Modernization	Green	-10	1	No Action
13	P009	E-Commerce Platform Replatfor	Completed	8	0	No Action
14	P010	Financial Close Automation	Green	-4	0	No Action
15	P014	Contact Center CRM Migration	Green	-9	2	No Action

10. Power BI Dashboard Walkthrough

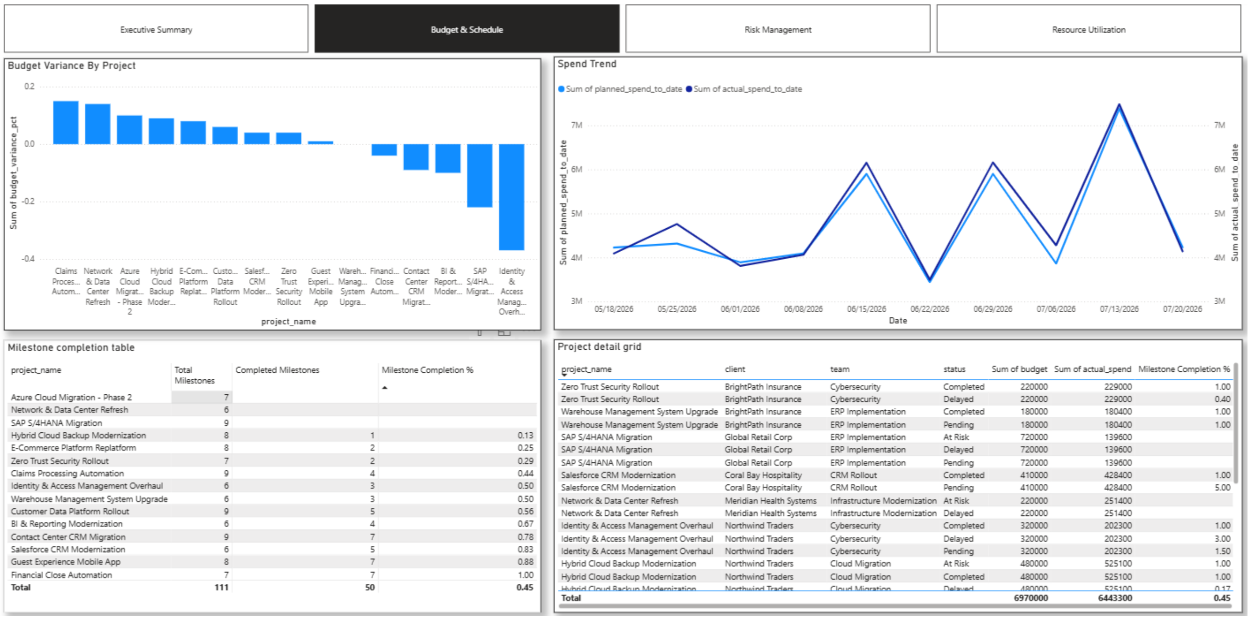
Page 1: Executive Summary



KPI cards for total projects, portfolio budget, actual spend, and variance %, a RAG-status donut, and a budget-vs-actual comparison by team.

7 of 15 projects (47%) are currently Red or Amber, 5 Amber and 2 Red, while 5 projects (33%) are Green and 3 (20%) are Completed. Despite this, the portfolio is running under budget overall (-7.56% variance), driven by heavy underspend in Cybersecurity and Cloud Migration offsetting overspend in ERP Implementation and Digital Transformation, both of which are tracking actual spend above budget. 11 high-impact risks remain open, a figure that warrants closer attention given the portfolio is not, in aggregate, a budget problem.

Page 2: Budget & Schedule



project_name	Total Milestones	Completed Milestones	Milestone Completion %
Azure Cloud Migration - Phase 2	7		
Network & Data Center Refresh	6		
SAP S/4HANA Migration	9		
Hybrid Cloud Backup Modernization	8	1	0.13
E-Commerce Platform Replatform	8	2	0.25
Zero Trust Security Rollout	7	2	0.29
Claims Processing Automation	9	4	0.44
Identity & Access Management Overhaul	6	3	0.50
Warehouse Management System Upgrade	6	3	0.50
Customer Data Platform Rollout	9	5	0.56
BI & Reporting Modernization	6	4	0.67
Contact Center CRM Migration	9	7	0.78
Salesforce CRM Modernization	6	5	0.83
Guest Experience Mobile App	8	7	0.88
Financial Close Automation	7	7	1.00
Total	**111**	**50**	**0.45**

project_name	client	team	status	Sum of budget	Sum of actual_spend	Milestone Completion %
Zero Trust Security Rollout	BrightPath Insurance	Cybersecurity	Completed	220000	229000	1.00
Zero Trust Security Rollout	BrightPath Insurance	Cybersecurity	Delayed	220000	229000	0.40
Warehouse Management System Upgrade	BrightPath Insurance	ERP Implementation	Completed	180000	180400	1.00
Warehouse Management System Upgrade	BrightPath Insurance	ERP Implementation	Pending	180000	180400	1.00
SAP S/4HANA Migration	Global Retail Corp	ERP Implementation	At Risk	720000	139600	
SAP S/4HANA Migration	Global Retail Corp	ERP Implementation	Delayed	720000	139600	
SAP S/4HANA Migration	Global Retail Corp	ERP Implementation	Pending	720000	139600	
Salesforce CRM Modernization	Coral Bay Hospitality	CRM Rollout	Completed	410000	428400	1.00
Salesforce CRM Modernization	Coral Bay Hospitality	CRM Rollout	Pending	410000	428400	5.00
Network & Data Center Refresh	Meridian Health Systems	Infrastructure Modernization	At Risk	220000	251400	
Network & Data Center Refresh	Meridian Health Systems	Infrastructure Modernization	Delayed	220000	251400	
Identity & Access Management Overhaul	Northwind Traders	Cybersecurity	Completed	320000	202300	1.00
Identity & Access Management Overhaul	Northwind Traders	Cybersecurity	Delayed	320000	202300	3.00
Identity & Access Management Overhaul	Northwind Traders	Cybersecurity	Pending	320000	202300	1.50
Hybrid Cloud Backup Modernization	Northwind Traders	Cloud Migration	At Risk	480000	525100	1.00
Hybrid Cloud Backup Modernization	Northwind Traders	Cloud Migration	Completed	480000	525100	1.00
Hybrid Cloud Backup Modernization	Northwind Traders	Cloud Migration	Delayed	480000	525100	0.17
Total				**6970000**	**6443300**	**0.45**

Weekly planned-vs-actual spend trend, milestone completion rate by project, and budget variance ranking.

Three projects: Azure Cloud Migration Phase 2, Network & Data Center Refresh, and SAP S/4HANA Migration, show zero completed milestones despite active budgets, and two of them are also running the highest cost overruns in the portfolio, meaning spend is outpacing plan with no delivery progress behind it. Portfolio-wide milestone completion sits at 45% (50 of 111), but that average hides a wide split, from fully complete projects down to these three stalled ones. Spend itself spikes sharply above plan in mid-June and again on July 13, reaching the portfolio's highest point (~7M) before dropping back, a pattern that points to lumpy, milestone-driven billing rather than steady burn.

Page 3: Risk Management

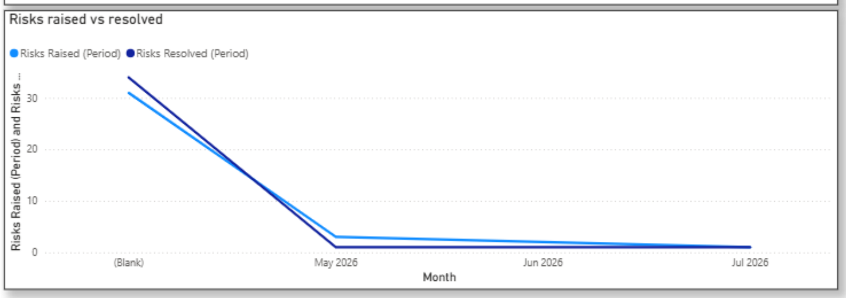
Executive Summary | Budget & Schedule | Risk Management | Resource Utilization

Top 10 open risks

project_name	description	likelihood Risk	impact on Project	Sum of risk_score
Azure Cloud Migration - Phase 2	Client sign-off on requirements delayed beyond baseline date	Medium	Medium	4
Azure Cloud Migration - Phase 2	Client-side change management readiness behind schedule	High	Medium	6
Azure Cloud Migration - Phase 2	Contractor visa/onboarding delay impacting staffing plan	Low	Medium	2
Azure Cloud Migration - Phase 2	Security review identified gaps in access control design	Low	High	3
BI & Reporting Modernization	Legacy data quality issues discovered during migration dry-run	High	High	9
Contact Center CRM Migration	Legacy data quality issues discovered during migration dry-run	High	High	9
Contact Center CRM Migration	Security review identified gaps in access control design	Low	High	3
Customer Data Platform Rollout	Client-side change management readiness behind schedule	High	Medium	6
Customer Data Platform Rollout	Interface defects found late due to incomplete test coverage	High	High	9
Customer Data Platform Rollout	Regulatory/compliance requirement changed mid-project	Low	High	3
Financial Close Automation	Cross-team dependency on shared integration environment	Medium	Medium	4
Total				**128**

Risk Heat Map

likelihood Risk	High	Low	Medium	**Total**
High	RSK0003		RSK0001	**RSK0001**
Low	RSK0004		RSK0017	**RSK0004**
Medium	RSK0009	RSK0028	RSK0002	**RSK0002**
Total	**RSK0003**	**RSK0028**	**RSK0001**	**RSK0001**



A likelihood-by-impact heat map, the top 10 open risks by score, and a raised-vs-resolved trend line.

The portfolio carries 128 open risk-score points, with risk concentrated in a small group of projects rather than evenly distributed across the portfolio. Azure Cloud Migration – Phase 2, Hybrid Cloud Backup Modernization, and Network & Data Center Refresh each have the highest number of open risks (4), followed by Customer Data Platform Rollout and Zero Trust Security Rollout with 3 each, making these projects the immediate focus for PMO risk review. The risk heat map also highlights a concentration of higher-impact risks, including security, data-quality, client-side change, and integration dependencies that could affect delivery timelines if not actively mitigated. Risk activity is heavily concentrated in May, when risk creation and resolution both peak, followed by a sharp decline through June and July. This suggests that the portfolio has moved past the initial risk-identification surge, but the remaining open risks are concentrated in strategically important projects and require active owners, mitigation plans, and escalation tracking to prevent them from becoming schedule or budget issues.

Page 4: Resource Utilization

name	team	role	Sum of allocation_pct
Omar Larsen	ERP Implementation	Project Manager	125.00%
Aisha Whitfield	Data & Analytics	UX Designer	100.00%
Ethan Chen	Cybersecurity	Developer	100.00%
Ethan Chen	Digital Transformation	Change Manager	100.00%
Ethan Larsen	Data & Analytics	Solution Architect	100.00%
Hana Chen	Data & Analytics	UX Designer	100.00%
Omar Adeyemi	Cloud Migration	Security Analyst	100.00%
Omar Ibrahim	Infrastructure Modernization	QA Engineer	100.00%
Omar Kowalski	CRM Rollout	Data Engineer	100.00%
Wei Tanaka	ERP Implementation	Integration Specialist	100.00%
Aisha Chen	Data & Analytics	Project Manager	75.00%
Aisha Whitfield	Cybersecurity	Solution Architect	75.00%
Chloe Dubois	ERP Implementation	Developer	75.00%
Chloe Novak	CRM Rollout	Solution Architect	75.00%
Emma Singh	ERP Implementation	Data Engineer	75.00%
Marco Kim	Digital Transformation	Security Analyst	75.00%
Marco Larsen	Data & Analytics	Project Manager	75.00%
Noah Singh	Digital Transformation	Project Manager	75.00%
Wei Fontaine	CRM Rollout	Security Analyst	75.00%
Aisha Chen	Cloud Migration	Project Manager	50.00%
Aisha Chen	Digital Transformation	Data Engineer	50.00%
Total			3750.00%

Total allocation % by resource against a 100% reference line, headcount by role, and team-level allocation load. Resource allocation is concentrated across a relatively small group of people and teams, with total portfolio allocation reaching 3,750%, indicating significant shared-resource dependency across projects. Several resources exceed the 100% capacity reference when their allocations are aggregated, with the highest allocation concentrated around Ethan Chen and Aisha Whitfield, creating potential over-allocation and delivery-risk concerns. Project Managers represent the largest role group at 15 resources (22% of headcount), followed by Data Engineers and Senior Developers with 7 each, showing that delivery leadership and technical execution capacity are the largest resource pools. At the team level, ERP Implementation carries the highest aggregate allocation load, followed by Data & Analytics and Digital Transformation, while Infrastructure Modernization and Cybersecurity operate with comparatively lower allocation levels. The concentration of resources across multiple teams suggests that PMO leadership should review capacity against project priorities, identify over-allocated individuals, and rebalance shared resources before capacity constraints begin affecting milestone delivery and project schedules.

11. Challenges & Problem-Solving

Genuine issues encountered during the build, and how each was diagnosed and resolved:

Problem	Root Cause	Resolution
RiskLog columns failed to query as expected	Source CSV headers contained spaces ("likelihood Risk", "impact on Project"), creating quoted identifiers in PostgreSQL	Renamed columns post-import with ALTER TABLE ... RENAME COLUMN to remove the need for quoting
ROUND (double precision, integer) does not exist	PostgreSQL has no two-argument ROUND () overload for double precision, only for numeric	Cast expressions to: numeric before rounding; corrected the underlying column type going forward
Date column silently misread or errored on import	Ambiguous day/month vs month/day format (e.g., 18/05/2026) parsed under the wrong locale	Used Power Query's "Data Type: Using Locale" option, explicitly set to the source's day/month/year format
USERELATIONSHIP () error in DAX measures	The function activates an existing relationship, it does not create one; no relationship existed yet between the two columns	Manually created both Calendar-to-RiskLog relationships in Model view before referencing them in DAX

This section is deliberately kept factual and specific, genuine debugging detail is what distinguishes hands-on build experience from a copied tutorial.

12. Business Impact & Estimated Value

Frame every metric here as decision-enabling, not just descriptive. Replace the placeholders below with your own honest estimates:

Reduced manual portfolio status compilation from an estimated 3–4 hours per week — typical for a PMO analyst manually pulling status across 15 projects and four tracking dimensions — to an automated, on-demand dashboard refresh.

Surfaced 18 resources currently allocated above 100% capacity across the portfolio, a resourcing risk that would otherwise require manually cross-referencing allocation spreadsheets project by project.

Identified 7 of 15 projects (47%) as Red or Amber and 3 projects with zero completed milestones despite active spend, cutting the time to flag at-risk projects ahead of Steering Committee from a manual review to under 30 seconds via the Executive Summary and Budget & Schedule pages.

Replaced subjective escalation judgment calls with a rule-based, auditable classification (via the Escalation Action logic) applied consistently across all 15 projects.

13. Skills Demonstrated

- SQL: filtering, joins, aggregation, CTEs, window functions (RANK, ROW_NUMBER, LAG), correlated subqueries
- Relational data modeling and schema design trade-offs
- Power BI: data modeling, relationship management, DAX (CALCULATE, SWITCH, USERELATIONSHIP, time intelligence)
- PMO domain knowledge: RAG reporting, risk scoring, resource utilization, escalation governance
- Systematic debugging and root-cause diagnosis across the database and BI layers
- Executive-facing dashboard design and stakeholder-ready reporting

14. Future Roadmap

Phase two of this project ("PMO Portfolio Tracker v2: Risk Prediction & Automated Escalation") extends the current model with a weighted risk-scoring algorithm and automated stakeholder alert logic, moving the project from descriptive reporting toward predictive PMO analytics.

15. Appendix

Full project files, including schema DDL, synthetic data CSVs, the complete 10-query SQL script, and the Power BI (.pbix) file, are available at:

https://github.com/Mayankchonde/Multi-client-PMO-portfolio-tracker